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Author contributions

E.A. and L.B trained the SAE models, built the InterProt visualizer, and interpreted SAE latents. E.A. led the feature characterization experiments. L.B led the interpretable probing experiments. M.L. wrote the code for SAE steering and performed the family-specific latent steering experiments. Y.Y. contributed to the training code. E.A., L.B., M.A., provided feedback on all experiments. E.A. and L.B. wrote the manuscript with feedback from M.L. and M.A. All authors read and approved the manuscript.

Competing interests

M.A. is a member of the scientific advisory boards of Cyrus Biotechnology, Deep Forest Sciences, Nabla Bio, and Oracle Therapeutics.

A. Appendix

A.1. Code availability and model weights

Our SAE training and evaluation code is publicly available at `https://github.com/etowahadams/interprot`. The feature visualization tool, *InterProt*, can be accessed at `https://interprot.com`, and pre-trained SAE weights are hosted at `https://huggingface.co/liambai/InterProt-ESM2-SAEs`.

A.2. Background on protein language models

Protein language models (pLMs) apply techniques from natural language processing to model biological sequences, leveraging the ability of transformer architectures to learn patterns from large-scale data. Just as language models capture structure in text, pLMs can learn meaningful representations of proteins from large protein sequence databases. The ESM family of pLMs are BERT-style transformers trained with a masked language modeling objective to learn contextual embeddings of amino acids (Rives et al., 2021; Lin et al., 2023). In this work, we focus on the 650M-parameter variant of ESM-2 (Lin et al., 2023) which has 33 total layers.

A.3. Key differences from InterPLM

Concurrent to our work, Simon & Zou (2024) also trained SAEs on ESM-2 models. Some key differences from work are as follows:

SAE Architecture. Simon & Zou (2024) used ReLU-based SAEs, we employ TopK SAEs, which allow explicit control over L0 sparsity.

Evaluation. Simon & Zou (2024) proposed an automated evaluation framework based on Swiss-Prot annotations, establishing that more SAE latents correspond to annotations compared to ESM neurons. We support this finding with a manual evaluation where participants blindly rated SAE latents and ESM neurons based on their interpretability in our visualization tool.

Analysis. Simon & Zou (2024) used an LLM to generate SAE feature descriptions based on activation examples and annotations. They also explored the application of SAE features in identifying missing annotations. Our analyses focused on classifying SAE latents based on activation patterns and family-specificity and using linear probes to uncover features most predictive in downstream tasks.

Visualization. Both works provide an interactive visualizer for SAE latents. interplm.ai from Simon & Zou (2024) display UMAP visualizations, results from Swiss-Prot concept mapping, and other statistics. Our visualizer, interprot.com, focus on the top activating sequences of each latent, displaying activation patterns overlaid on structure, sequence alignments, and information on shared protein family. We used this interface to conduct our human ratings experiment.

A.4. Probe details

Table 1 summarizes the linear probes used in each downstream task.

Table 1. Summary of downstream tasks for interpretable probing

Dataset	Task type	Label level	Evaluation metric	Probe implementation
Secondary structure	Classification	Residue	Accuracy	PyTorch linear classifier
Subcellular location	Classification	Protein	Accuracy	Sklearn logistic regression
Thermostability	Regression	Protein	Spearman's ρ	Sklearn ridge regression
CHO cell expression	Classification	Protein	Accuracy	Sklearn logistic regression